

KARNATAKA 2ND PUC BOARD IMPORTANT QUESTIONS & HIGH-YIELD TOPICS

Chapter 4: Principles of Inheritance and Variation (Subject Code: 36 • Weightage: 13 Marks)

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1. OFFICIAL KARNATAKA 2ND PUC BLUEPRINT ALLOTMENT

Chapter Name & Code	Periods Allotted	Official Weightage	Cognitive Domain Distribution (Blueprint 2024–25)
Ch 4: Principles of Inheritance & Variation	15 Hours	13 Marks	• Application (8 Marks): 1 MCQ (1M) + 1 SA2 (2M) + 1 LA (5M) • HOTS (5 Marks): 1 LA (5M)

2. KEY TOPICS TO BE FOCUSED ON (HIGH-YIELD MASTER CHECKLIST)

Priority Focus Areas for Maximum Board Scoring

• Top 5-Mark Topics (Part D - Sec V & VI):

1. **Dihybrid Cross & Law of Independent Assortment:** Full cross ($RRYY \times rryy$), 16-box Punnett square, 9:3:3:1 phenotypic & 1:2:1:2:4:2:1:2:1 genotypic ratios, and statement of the law.
2. **ABO Blood Grouping:** Multiple allelism (I^A, I^B, i) & co-dominance ($I^A I^B$), complete 6 genotypes & 4 phenotypes table.
3. **Chromosomal Disorders:** Down's (47, +21), Klinefelter's (47, XXY), and Turner's (45, X0) syndromes with karyotypes and distinct clinical symptoms.
4. **Sickle Cell Anaemia (HOTS):** Molecular basis ($GAG \rightarrow GUG$; Glu $\rightarrow$ Val at pos 6 of β -chain on Chr 11) and probability cross between carrier parents.

• **Core 2-Mark Topics (Part B):** Test cross (definition, cross, 1:1 ratio), Incomplete Dominance vs. Co-dominance, Morgan's reasons for selecting *Drosophila*, Pleiotropy (PKU), and Thalassemia (α vs β).

• **Objective 1-Mark Topics (Part A):** Sex determination systems (Honeybee haplodiploidy, Bird ZW-ZZ), Sturtevant's map unit (cM), and Morgan's linkage values (1.3% vs 37.2%).

3. PART A: 1-MARK HIGH-YIELD QUESTIONS (MCQS & OBJECTIVE)

Q1 In snapdragon (*Antirrhinum majus*), a cross between red-flowered and white-flowered plants produces pink-flowered F_1 plants. What is the phenotypic ratio in F_2 ?

1 Mark (Part A)

Answer: 1 Red : 2 Pink : 1 White (1:2:1) due to incomplete dominance.

Q2 Name the type of sex determination observed in honeybees.

1 Mark (Part A)

Answer: Haplodiploid sex determination (Females $2n = 32$, Males $n = 16$).

Q3 What is the karyotype and cause of Klinefelter's syndrome?

1 Mark (Part A)

Answer: 47, XXY (Allosomal trisomy due to presence of an extra X chromosome in males).

Q4 Which unit is used to measure the physical distance between two genes on a chromosome?

1 Mark (Part A)

Answer: Centimorgan (cM) or Map unit (m.u.) (1% recombination = 1 map unit).

4. PART B: 2-MARK IMPORTANT QUESTIONS (SHORT ANSWERS)

Q5 What is a test cross? Mention its genetic significance.

2 Marks (Part B)

Answer:

- **Definition:** A cross between an individual of unknown dominant phenotype with a homozygous recessive parent ($_? \times tt$).
- **Significance:** Used to determine whether the dominant phenotype is homozygous dominant ($TT \rightarrow$ yields 100% tall) or heterozygous ($Tt \rightarrow$ yields 1:1 ratio).

Q6 Differentiate between Incomplete Dominance and Co-dominance with one example each.

2 Marks (Part B)

Answer:

- **Incomplete Dominance:** The F_1 phenotype is an intermediate blend between both parents (e.g., Pink flowers in *Antirrhinum majus*, Rr).
- **Co-dominance:** Both alleles express themselves equally and independently in heterozygotes without blending (e.g., AB blood group in humans, $I^A I^B$).

Q7 State two reasons why Morgan selected *Drosophila melanogaster* for his genetic experiments.

2 Marks (Part B)

Answer:

1. They can be grown easily on simple synthetic medium in the laboratory and complete their life cycle in just 2 weeks.
2. Clear sexual dimorphism (male and female easily distinguishable) and a single mating produces a large number of progeny.

Q8 What is pleiotropy? Give one example in humans.

2 Marks (Part B)

Answer:

- **Definition:** The phenomenon where a single gene exhibits multiple phenotypic expressions by affecting metabolic pathways.
- **Example:** Phenylketonuria (PKU) — mutation in phenylalanine hydroxylase gene causes mental retardation, reduction in hair, and skin pigmentation.

5. PART D (SEC V & VI): 5-MARK HIGH-YIELD & HOTS QUESTIONS

Q9 Explain the inheritance of two genes (Dihybrid Cross) in *Pisum sativum* with a Punnett Square and state the Law of Independent Assortment.

5 Marks (Part D - Sec V)

Model Answer:

1. **Cross Details:** Homozygous round yellow seeded plant ($RRYY$) crossed with homozygous wrinkled green seeded plant ($rryy$).
2. **F_1 Generation:** Produces heterozygous Round Yellow seeds ($RrYy$).
3. **F_2 Punnett Square:** Selfing $RrYy \times RrYy$ produces 4 types of gametes (RY, Ry, rY, ry) yielding 16 combinations.
4. **F_2 Ratios:**
 - **Phenotypic Ratio:** 9 ext{ Round Yellow} : 3 ext{ Round Green} : 3 ext{ Wrinkled Yellow} : 1 ext{ Wrinkled Green} (9:3:3:1).
 - **Genotypic Ratio:** 1:2:1:2:4:2:1:2:1.
5. **Law of Independent Assortment:** "When two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair during gamete formation."

Q10 Describe the genetic basis of ABO Blood Groups in human beings. Explain how it exemplifies both Co-dominance and Multiple Allelism.

5 Marks (Part D - Sec V)

Model Answer:

1. **Gene & Alleles:** ABO blood groups are controlled by gene I which has 3 alleles: I^A, I^B, i .
2. **Function:** I^A produces sugar A, I^B produces sugar B, while allele i does not produce any surface antigen sugar.
3. **Dominance Relationships:** I^A and I^B are completely dominant over allele i .
4. **Co-dominance:** When I^A and I^B are present together in genotype $I^A I^B$, both express their respective sugars independently resulting in blood group AB.
5. **Genotypes & Phenotypes Table:** 6 genotypes ($I^A I^A, I^A i, I^B I^B, I^B i, I^A I^B, ii$) coding for 4 phenotypes (A, B, AB, O).

Q11 (a) What is sickle cell anaemia? Explain its molecular basis and inheritance. (b) A carrier male marries a carrier female. Work out the probability of their children being diseased. **5 Marks (HOTS / Part D - Sec VI)**

Model Answer:

1. **Molecular Cause:** Autosomal recessive disorder caused by a single base substitution (transversion) in the β -globin gene on Chromosome 11 from *GAG* $\rightarrow$ *GUG*. This causes substitution of Glutamic acid (Glu) by Valine (Val) at the 6th position of the β -globin chain.

2. **Pathophysiology:** Mutant haemoglobin (Hb^S) polymerizes under low oxygen tension, changing biconcave RBCs into rigid sickle shapes, blocking capillaries.

3. **Genetics & Cross:**

- Parents: Father ($Hb^A Hb^S$) $\times$ Mother ($Hb^A Hb^S$)
- Offspring genotypes: $1 \text{ } Hb^A Hb^A \text{ (Normal)} : 2 \text{ } Hb^A Hb^S \text{ (Carrier)} : 1 \text{ } Hb^S Hb^S \text{ (Diseased)}$.
- Probability of diseased child: 25% or $\frac{1}{4}$.

Q12 Explain the cause, karyotype, and any three salient symptoms of each of the following: **5 Marks (Part D - Sec V / VI)**
(a) Down's Syndrome (b) Turner's Syndrome.

Model Answer:

• (a) **Down's Syndrome:**

1. **Cause & Karyotype:** Autosomal trisomy of chromosome 21 ($47, XX/XY +21$) due to non-disjunction.
2. **Symptoms:** Short stature, small round head, furrowed tongue with partially open mouth, characteristic palm crease, congenital heart defects, and retarded psychomotor/mental development.

• (b) **Turner's Syndrome:**

1. **Cause & Karyotype:** Autosomal monosomy due to absence of one X chromosome ($45, X0$).
2. **Symptoms:** Sterile female, rudimentary streak ovaries, short stature, webbed neck, broad shield-shaped chest with widely spaced nipples, lack of secondary sexual characters, and amenorrhea.

Narendra Sir's Blueprint Scoring Tips (13/13 Marks Target)

1. **Mandatory 5-Mark Visuals:** Always practice drawing the Dihybrid 16-square Punnett square and the Pedigree charts for Myotonic dystrophy & Sickle cell anaemia.
2. **HOTS Analytical Focus:** Master pedigree chart interpretation and blood group parentage crosses (I^A, I^B, i).
3. **Syndromes:** Clearly memorize Karyotypes ($47, +21; 47, XXY; 45, X0$) and exact textbook clinical features.